LSTM-Based Sentiment Analysis on IMDb Movie Reviews

Ali Hamza

Department of Computer Science and Artificial Intelligence, PAF-IAST, Haripur, Pakistan

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Abstract: This report presents an LSTM-based approach for binary sentiment classification on the IMDb movie review dataset. The study covers dataset preprocessing, sequence generation, baseline model development, performance evaluation, and parameter modification for improved generalization. A baseline LSTM model achieved a test accuracy of 0.8164, while a modified configuration with 128 hidden units and dropout 0.3 improved performance to 0.8318. The results show that sequence-aware neural architectures are effective for sentiment analysis tasks in which contextual order and negation patterns strongly affect semantic interpretation.

Index Terms: LSTM, sentiment analysis, IMDb dataset, natural language processing, sequence modeling, binary classification.

I. INTRODUCTION

Sentiment analysis is a fundamental natural language processing task in which a model must determine whether a text expresses positive or negative opinion. In the present lab, the target application is the automatic classification of IMDb movie reviews into binary sentiment categories. This problem is not ideally handled by fixed-input multilayer perceptrons because sentiment often depends on local word order and long-range contextual relationships.

Long Short-Term Memory networks are well suited to this scenario because they process text as an ordered sequence and preserve relevant information through gated memory. Consequently, they can better interpret expressions such as `not good`, where polarity is determined by interaction between multiple tokens rather than isolated word presence.

The objectives of this lab were to preprocess the IMDb 50K review dataset, construct a baseline LSTM model, evaluate its learning behavior, modify critical parameters, and compare the modified system against the original baseline in terms of validation and test performance.

II. METHODOLOGY

A. Dataset Preparation

The dataset used in this experiment consists of 50,000 labeled movie reviews. Before training, the raw text was cleaned by removing HTML artifacts, normalizing whitespace, lowercasing the text, and removing unwanted characters. The cleaned reviews were then tokenized and converted into fixed-length integer sequences for input into the LSTM model.

TABLE I
PREPROCESSING CONFIGURATION

Component	**Value / Method**
Dataset Source	IMDB Dataset.csv with 50,000 movie reviews
Vocabulary Size	Tokenizer(num_words=5000, oov_token="<OOV>")
Sequence Length	100 tokens
Data Split	Train / Validation / Test = 40000 / 5000 / 5000
Padding Strategy	padding="post", truncating="post"
Label Encoding	Negative = 0, Positive = 1

LISTING 1. TEXT CLEANING AND SEQUENCE PREPARATION

```
def clean_review(text):
    text = html.unescape(str(text))
    text = re.sub(r"<[^>]+>", " ", text)
    text = text.replace("\n", " ")
    text = re.sub(r"[^a-zA-Z0-9' ]+", " ", text)
    text = re.sub(r"\s+", " ", text).strip().lower()
    return text

tokenizer = Tokenizer(num_words=5000, oov_token="<OOV>")
tokenizer.fit_on_texts(train_df["review_clean"])

X_train = pad_sequences(train_sequences, maxlen=100,
                        padding="post", truncating="post")
```

Padding is necessary because text samples naturally vary in length, whereas mini-batch training requires all inputs in a batch to share a common tensor shape. By selecting a fixed sequence length of 100, the model receives a consistent amount of context while maintaining computational efficiency.

Task 1: Sample IMDb Reviews

Sentiment: Positive

i really liked this summerslam due to the look of the arena the curtains and just the look overall was interesting to me for some reason anyways this could have been one of the best summerslam's ever if the wwf didn't have lex luger in the main event against yokozuna now for it's time it was ok to have a huge fat man vs a strong man but i'm glad times have changed it was a terrible main event just like every ...

Sentiment: Positive

not many television shows appeal to quite as many different kinds of fans like farscape does i know youngsters and 30 40 years old fans both male and female in as many different countries as you can think of that just adore this t v miniseries it has elements that can be found in almost every other show on t v character driven drama that could be from an australian soap opera yet in the same episode it has ...

Sentiment: Negative

the film quickly gets to a major chase scene with ever increasing destruction the first really bad thing is the guy hijacking steven seagal would have been beaten to pulp by seagal's driving but that probably would have ended the whole premise for the movie it seems like they decided to make all kinds of changes in the movie plot so just plan to enjoy the action and do not expect a coherent plot turn any sense of ...

Sentiment: Positive

jane austen would definitely approve of this one gwyneth paltrow does an awesome job capturing the attitude of emma she is funny without being excessively silly yet elegant she puts on a very convincing british accent not being british myself maybe i'm not the best judge but she fooled me she was also excellent in sliding doors i sometimes forget she's american also brilliant are jeremy northam and sophie ...

Sentiment: Negative

expectations were somewhat high for me when i went to see this movie after all i thought steve carell could do no wrong coming off of great movies like anchorman the 40 year old virgin and little miss sunshine boy was i wrong i'll start with what is right with this movie at certain points steve carell is allowed to be steve carell there are a handful of moments in the film that made me laugh and it's due almost ...

Fig. 1. Sample cleaned reviews used to verify proper normalization before tokenization.

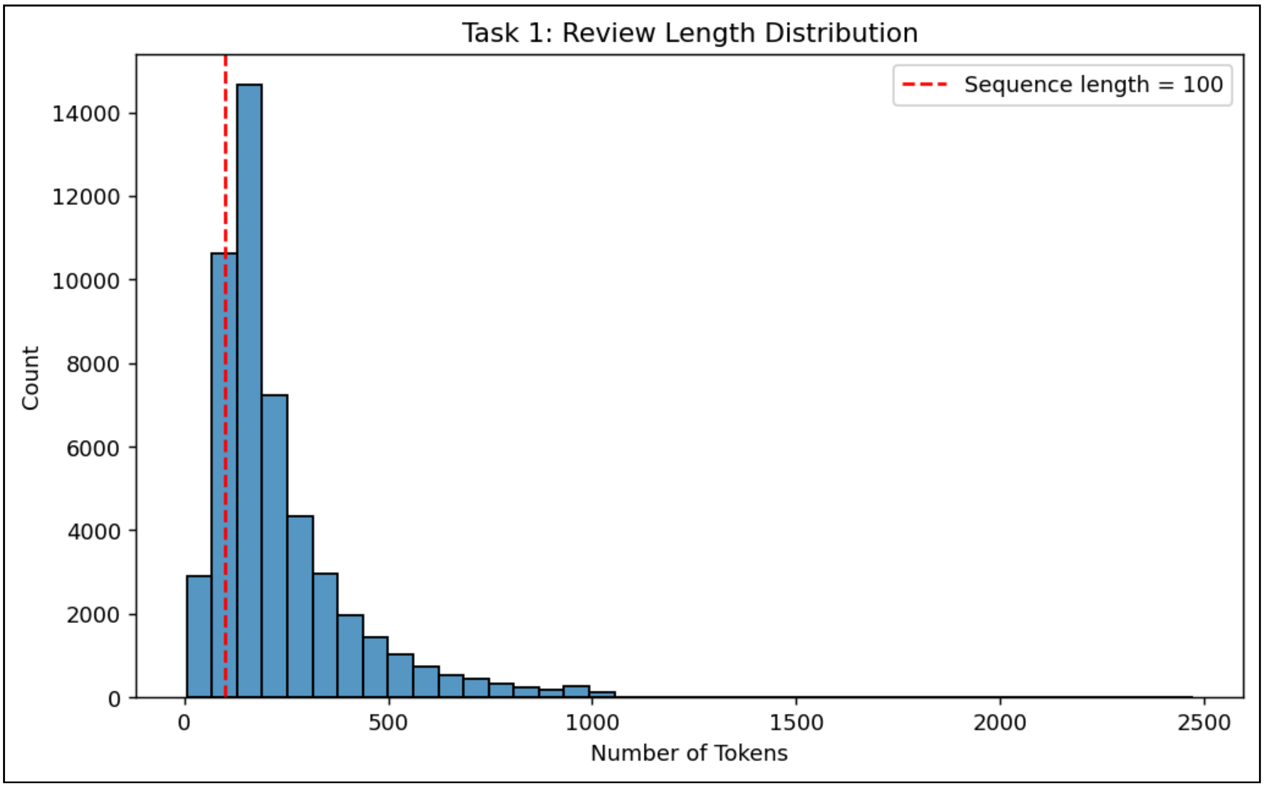

Fig. 2. Distribution of tokenized review lengths with the selected maximum length of 100 tokens.

B. Baseline LSTM Architecture

The baseline model consists of an Embedding layer, a single LSTM layer with 64 units, and a sigmoid output neuron. The Embedding layer converts token indices into dense vector representations, the LSTM extracts sequential dependencies, and the sigmoid output produces a binary sentiment probability.

LISTING 2. BASELINE AND REUSABLE LSTM MODEL DEFINITIONS

```
model = tf.keras.Sequential([
    tf.keras.layers.Embedding(input_dim=5000,
                              output_dim=64,
                              input_length=100),
    tf.keras.layers.LSTM(64),
    tf.keras.layers.Dense(1, activation='sigmoid')
])

def build_lstm_model(max_len, lstm_units=64,
                     dropout_rate=0.0, learning_rate=1e-3):
    model = tf.keras.Sequential([
        tf.keras.layers.Embedding(input_dim=MAX_WORDS,
                                  output_dim=EMBEDDING_DIM,
                                  input_length=max_len),
        tf.keras.layers.LSTM(lstm_units, dropout=dropout_rate),
        tf.keras.layers.Dense(1, activation="sigmoid"),
    ])
    model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate),
                  loss="binary_crossentropy",
                  metrics=["accuracy"])
    return model
```

This architecture was selected because sequence learning is essential for sentiment analysis. In contrast to SimpleRNN, the LSTM uses gated memory to retain important signals over longer spans and therefore better captures contextual reversals and sentiment-bearing phrases.

III. EXPERIMENTAL RESULTS

A. Baseline Training Performance

The original model was trained for five epochs with a batch size of 32. The resulting metrics indicate that the model learned the training data effectively but exhibited a noticeable generalization gap on the validation set.

Training Accuracy: 0.8817

Validation Accuracy: 0.8142

Training Loss: 0.2873

Validation Loss: 0.4262

Test Accuracy: 0.8164

Test Loss: 0.4230

LISTING 3. BASELINE TRAINING PROCEDURE

```
history_original = original_model.fit(
    X_train_original,
    y_train,
    validation_data=(X_val_original, y_val),
    epochs=5,
    batch_size=32,
    verbose=1,
)

original_test_loss, original_test_acc = original_model.evaluate(
    X_test_original, y_test, verbose=1
)
```

The difference between training and validation performance indicates mild overfitting. The model captured the training distribution more strongly than the held-out data, suggesting a need for additional regularization or architectural adjustment.

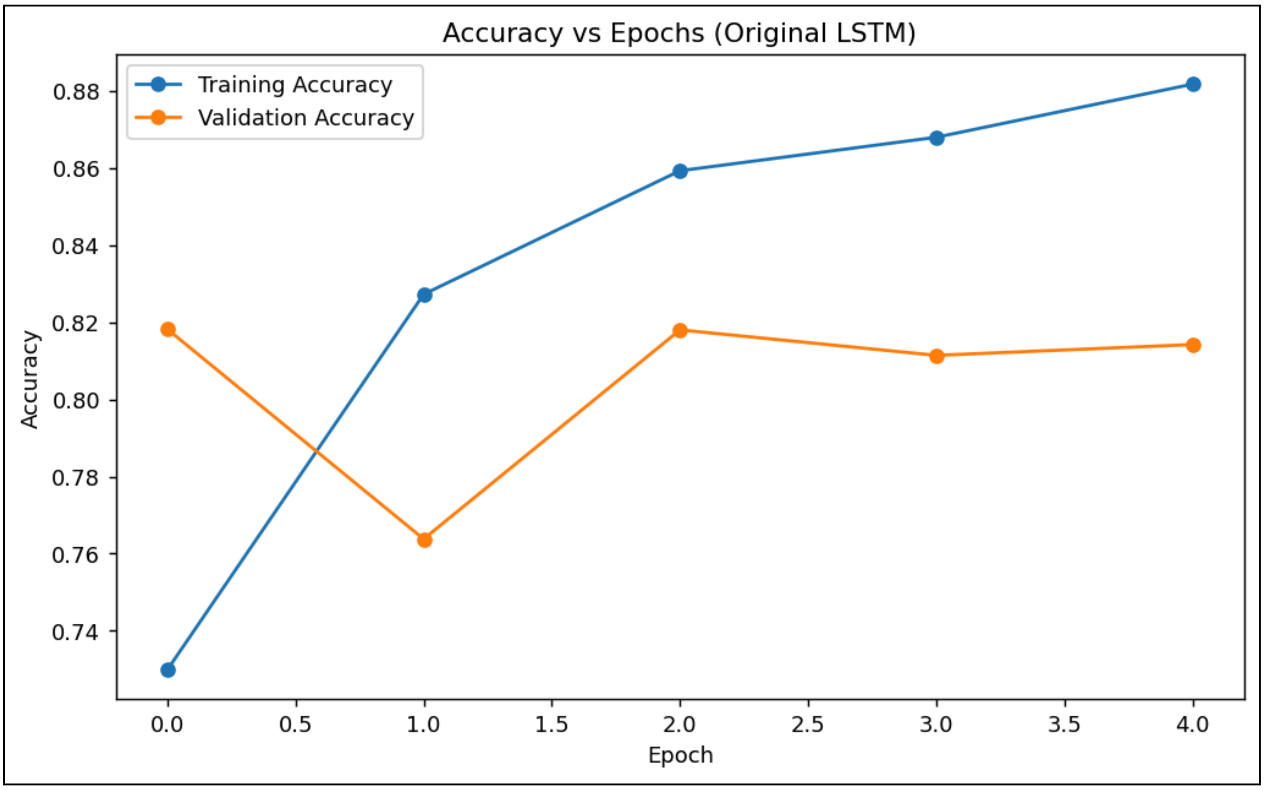

Fig. 3. Accuracy curves for the original LSTM model across training epochs.

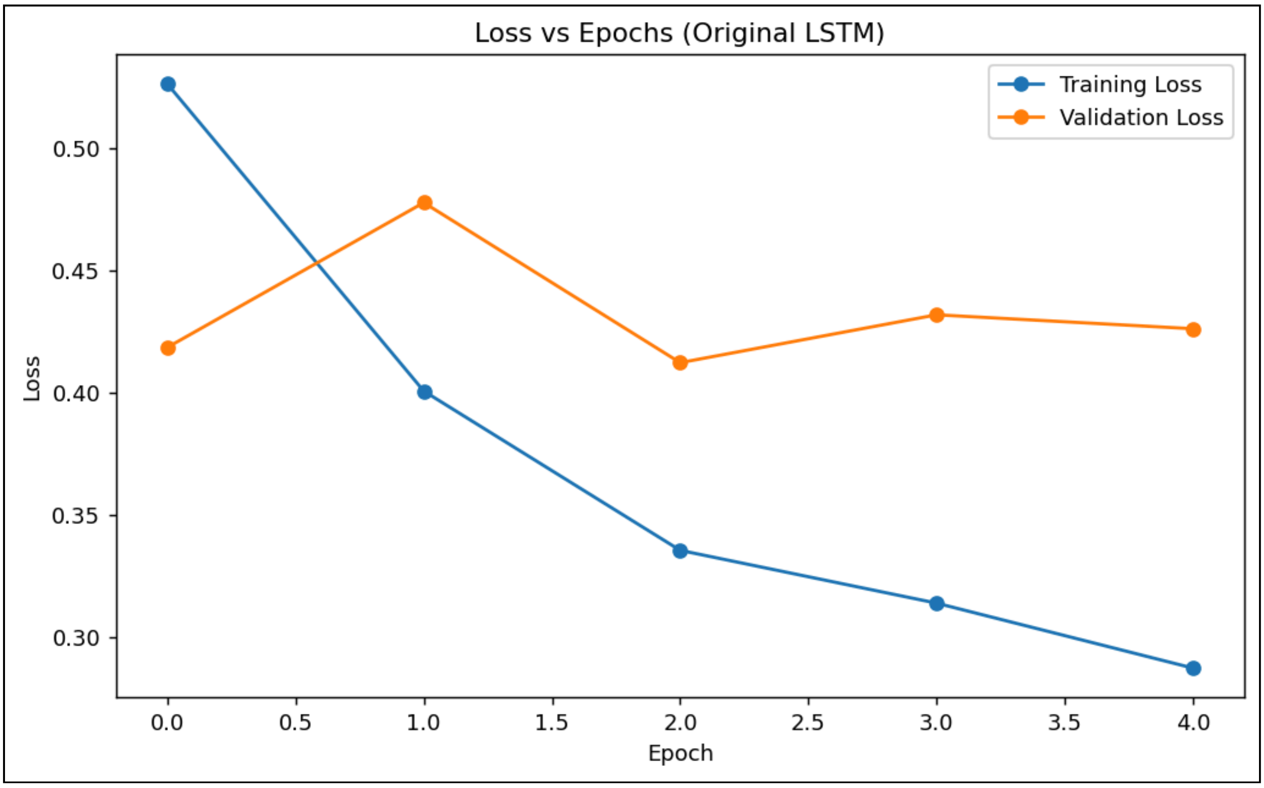

Fig. 4. Loss curves for the original LSTM model. Validation loss remains higher than training loss.

B. Modified Model and Comparative Analysis

To improve performance, the LSTM size was increased from 64 to 128 units and dropout of 0.3 was applied. The intention behind these changes was to improve representational capacity while counteracting overfitting through regularization.

LISTING 4. MODIFIED MODEL CONFIGURATION

```
modified_model = build_lstm_model(
    max_len=100,
    lstm_units=128,
    dropout_rate=0.3,
)
```

TABLE II
COMPARISON OF ORIGINAL AND MODIFIED MODELS

Model	**Train Acc.**	**Val. Acc.**	**Test Acc.**	**Train Loss**	**Val. Loss**	**Test Loss**
Original	0.8817	0.8142	0.8164	0.2873	0.4262	0.4230
Modified	0.8826	0.8318	0.8318	0.2818	0.3979	0.4001

The modified model improved validation accuracy from 0.8142 to 0.8318, which represents an absolute improvement of 0.0176. Test accuracy increased by the same magnitude, confirming that the modification improved out-of-sample generalization rather than merely fitting the training data more aggressively.

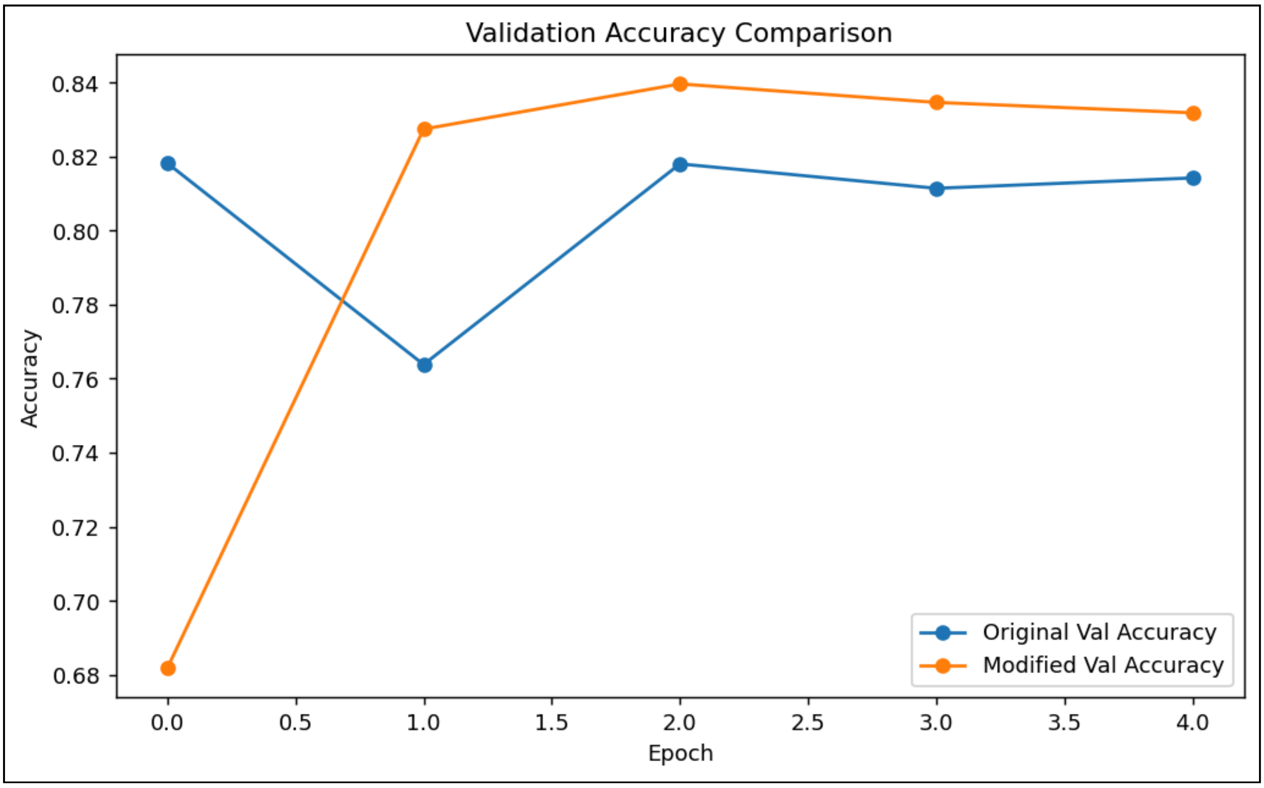

Fig. 5. Validation accuracy comparison between the baseline and modified LSTM configurations.

C. Final Prediction Analysis

Because the modified model achieved the best validation performance, it was selected as the final model for custom review prediction and final test-set analysis.

TABLE III
CUSTOM REVIEW PREDICTIONS USING THE FINAL MODEL

Review	**Predicted Probability**	**Predicted Sentiment**
This movie was absolutely amazing	0.9819	Positive
This movie was not good at all	0.2619	Negative
The acting was great but the story was too slow and boring	0.0548	Negative
I really loved the characters and the ending was satisfying	0.9863	Positive
I expected something better, and honestly it was a waste of time	0.0165	Negative

These outputs show that the final model reliably classifies strongly positive and strongly negative reviews, while also handling mixed expressions in which negative wording dominates the final interpretation.

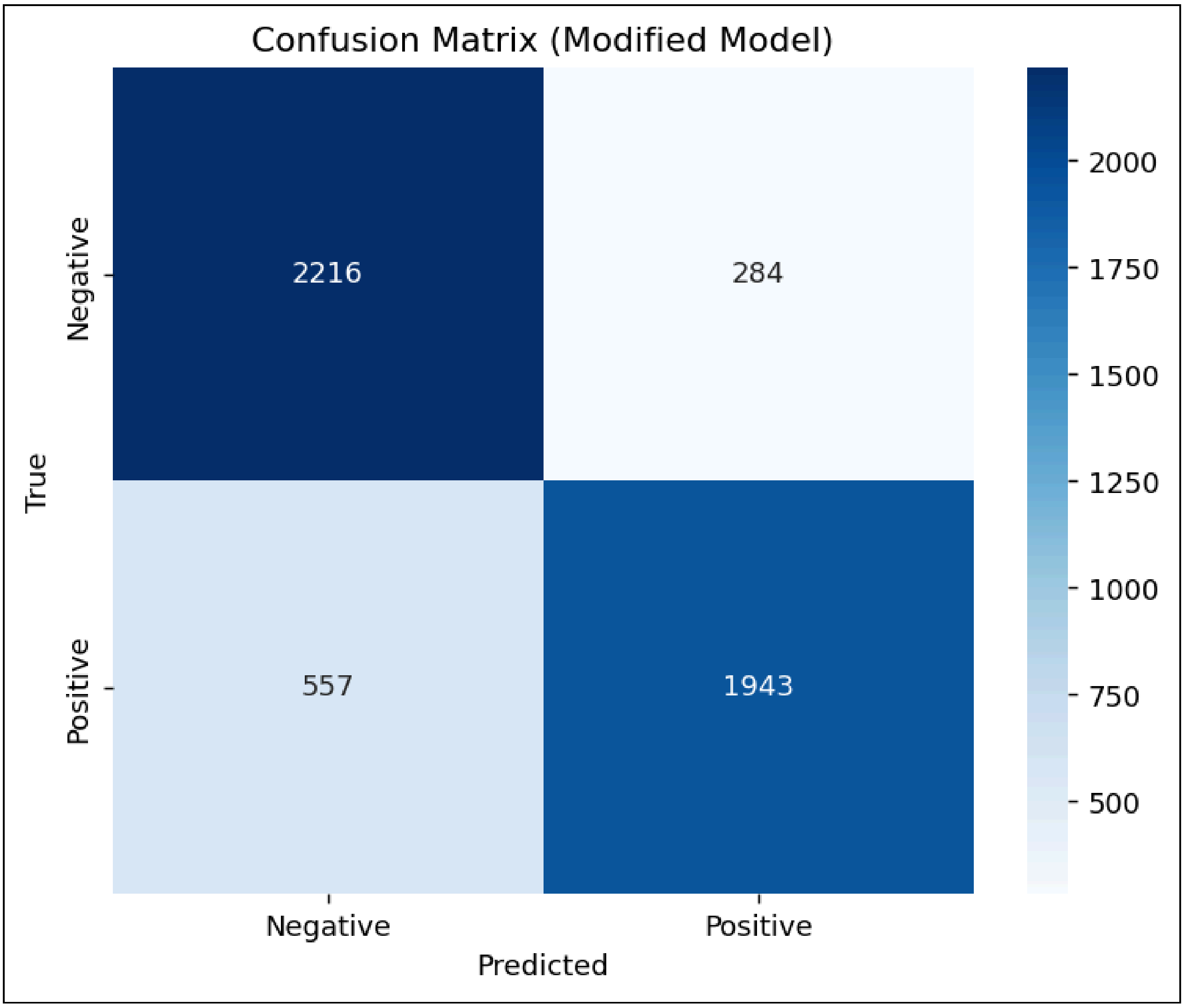

Fig. 6. Confusion matrix of the final modified model on the held-out test set.

LISTING 5. CLASSIFICATION REPORT OF THE FINAL MODEL

```
              precision    recall  f1-score   support

    Negative     0.7991    0.8864    0.8405      2500
    Positive     0.8725    0.7772    0.8221      2500

    accuracy                         0.8318      5000
   macro avg     0.8358    0.8318    0.8313      5000
weighted avg     0.8358    0.8318    0.8313      5000
```

The report shows stronger recall for the negative class than for the positive class. This indicates that, in the present experiment, the model was slightly more effective at detecting negative sentiment.

IV. DISCUSSION

A. Why LSTM Outperforms SimpleRNN

LSTM performs better than SimpleRNN because it uses input, forget, and output gates to manage the flow of information through time. This design mitigates the vanishing-gradient problem and enables the model to retain useful contextual cues over longer sequences. As a result, it is more effective for sentiment analysis, where polarity may depend on interactions among multiple words.

B. Role of the Memory Cell

The memory cell acts as a persistent state that carries relevant information across time steps. This allows the model to connect earlier expressions with later tokens and interpret phrases such as `not good` correctly instead of treating each word independently.

C. Importance of Padding and Sequence Length

Padding is required because batch-based deep learning methods expect inputs with identical shape. At the same time, sequence length must be chosen carefully. If the selected maximum length is too small, important sentiment-bearing tokens may be truncated; if it is too large, training becomes more expensive and may introduce unnecessary noise.

D. Effect of Increasing LSTM Units

Increasing the number of LSTM units improves representational capacity by allowing the model to learn a richer set of sequential features. In this lab, expanding the hidden state to 128 units and adding dropout produced a better balance between learning capacity and regularization, which improved both validation and test accuracy.

V. CONCLUSION

This lab demonstrated the complete workflow for LSTM-based sentiment analysis on the IMDb review dataset, including preprocessing, sequence construction, baseline modeling, evaluation, and architecture refinement. The baseline model performed well but showed mild overfitting, which motivated a modified design with a larger hidden state and dropout regularization.

The modified model achieved a final test accuracy of 83.18%, outperforming the original baseline and confirming that careful parameter tuning can improve generalization in sequence classification problems. Overall, the experiment reinforces the suitability of LSTM architectures for tasks in which contextual order and phrase-level dependencies determine sentiment.

VI. REFERENCES

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